

Ethnicity Classification using Deep Learning

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ABSTRACT

This study explores ethnicity classification using deep learning, with applications in identity verification, demographic analysis, and bias mitigation in artificial intelligence systems. It aims to develop a model that accurately classifies ethnicities in Malaysia, focusing on major ethnic groups such as Malay, Chinese, and Indian. The research uses a Convolutional Neural Network based on the VGG-Face architecture and a custom dataset reflecting Malaysia's ethnic diversity to address limitations in existing global datasets. Key findings suggest that the proposed model can accurately classify ethnicities, contributing to better demographic analysis and data collection. This research highlights the potential of deep learning to enhance service delivery and governance in Malaysia, particularly in areas like healthcare and social services, where ethnicity-based data can help optimize the distribution of resources and ensure fair access to services.

Keywords: Artificial Intelligence, Classification Model, Deep Learning, Data Collection, Ethnicity

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INTRODUCTION

Malaysia's multicultural population with major ethnic groups like Malay, Chinese and Indian requires accurate, large-scale demographic analysis to guide fair resource allocation. Manual or self-reported ethnicity labels are slow, labour-intensive and prone to bias, underscoring the need for automated solutions. Despite advancements in deep learning, existing models often struggle to classify Malaysia's ethnic groups accurately due to limited representation in global datasets. These datasets, focused mainly on Western populations, fail to capture the unique features of Malaysia's diverse demographics. Therefore, there is a critical need for more localized datasets and tailored approaches that accurately reflect Malaysia's diverse ethnic composition, ensuring more precise and inclusive results

Deep learning has seen significant progress, especially through Convolutional Neural Networks (CNNs), which are highly effective at capturing layered image features. This capability has led to remarkable performance in applications such as object detection and facial recognition (Alzubaidi et al., 2021). The study aims to develop an ethnicity classification system using a CNN-based deep learning model trained on a dataset curated to represent Malaysia's major ethnic groups and deliver reliable results. It explores how factors such as dataset diversity and model architecture impact classification accuracy and reliability. This work hopes to advance demographic studies and support inclusivity through technology tailored to Malaysia's unique population.

LITERATURE REVIEW

Deep learning frameworks are fundamental for designing, training, and evaluating neural networks, particularly for image-based tasks like ethnicity classification. Among the most widely adopted are TensorFlow (with the Keras API) and PyTorch. TensorFlow, especially when used with Keras, provides a high-level, user-friendly interface that simplifies model development. According to Chicho and Sallow (2021), Keras abstracts much of TensorFlow's complexity, enabling rapid prototyping and deployment, making it ideal for applied research and users with limited experience. Although PyTorch is praised for its flexibility and execution speed due to its

dynamic computation graph (Novac et al., 2022), this study adopts TensorFlow/Keras for its balance of ease of use, accuracy, and suitability for rapid experimentation.

Convolutional Neural Networks (CNNs) have transformed image classification tasks such as ethnicity classification by effectively learning spatial feature hierarchies automatically. Various CNN architectures have been developed and tested for ethnicity classification, each with its strengths and limitations. The overview of the CNN architectures used in previous research studies are shown in Table 1.

Table 1. Summary of previous studies on CNN architectures

Author(s)	Model	Accuracy	Ethnic Groups	Advantages	Disadvantages
Wang et al. (2016)	5-layer CNN	99.5%	Han, Uyghur, Non-Chinese	Automatically learns effective feature representations	Struggles with variations in lighting and head pose
Vo et al. (2018)	RR-CNN, RR-VGG	88.64%, 88.87%	Vietnamese, Other	Lightweight and effective for specific ethnic groups	Limited scope (only "Vietnamese" and "Other")
Jilani et al. (2019)	ResNet-50, ResNet-101, ResNet-152	98.8%, 99.2%, 99%	Pakistani	Deep residual learning improves accuracy and reduces computation time	Deeper networks require more computational resources
Ahmed et al. (2020)	R-Net (9-layer CNN)	77%, 25%, 80%, 79%	Caucasian, African, Asian, Indian	Effectively extracts features from images	Limited by computational resources; may overfit
Ache et al. (2023)	VGG16 + SVM	96%	Iranian, Asian	Improved feature extraction and reliable predictions	Computationally intensive; requires careful parameter tuning
Alghaili et al. (2023)	CNN-NN4 + SVM	97.3%	Chinese, Pakistani, Russian	Uses KL loss for effective classification with limited images	Complexity in loss function integration complicates training
Jahan & Mustafa (2024)	Customized DenseNet121	94%	Chakma, Marma, Tripura	Promotes feature reuse and efficient parameter use	Complex implementation; resource-intensive
Kalkatawi & Saeed (2024)	MaxViT (Transformer-based)	77.2%	Asian, Black, Indian, Latino, Middle Eastern, White	Better generalization; lower computational requirements	More resources and understanding than simpler CNNs

Transfer learning has also proven highly effective for ethnicity classification, especially when working with small or imbalanced datasets. Utilizing pre-trained models like EfficientNetB3, VGG-16, and VGG-Face that were originally trained on extensive datasets such as ImageNet can help model to achieve better generalization across different ethnic groups with less training data (Makolo et al., 2023).

The success of ethnicity classification models is largely influenced by the availability of large and diverse datasets. Several datasets have been developed and made public to support this research area, each with varying coverage of ethnic groups. Some of the common datasets used is presented in Table 2.

Table 2. Summary of datasets used in previous studies

Dataset Name	Year	Race/Ethnicity	Images	Sources
UTKFace	2017	Asian, Black, Indian, White, Others (Hispanic, Latino, Middle Eastern)	20,000	MORPH, CACD, online resources
IJB-C	2018	North & South America, Western Europe, Southwest Africa, East Europe, East Africa-Middle East, Southeast & East Asia, India, China	31,334	Public, law enforce databases, social media

IISCIFD	2019	North Indian, South Indian	1,647	Volunteers, Internet
BUPT-Balanced	2020	African, Asian, Caucasian, Indian	1.3 M	MS-Celeb
FairFace	2021	Black, East Asian, Indian, Latino Hispanic, Middle Eastern, Southeast Asian, White	108,192	Flickr, Twitter, news, online sources
DiveFace	2021	East Asia (Japan, China, Korea), Europe, North America, Latin America, Sub-Saharan Africa, India, South Asia	120,000	MegaFace

METHODOLOGY

The general architecture for ethnicity classification uses Convolutional Neural Networks (CNNs), which are based on simple neural networks. Figure 1 illustrates a CNN structure used for this task. It starts with an input layer that takes in the raw image. The image then goes through a several convolutional layers, where kernels (filters) are applied to recognize key features like edges, textures, and facial details. After each convolutional layer, a Rectified Linear Unit (ReLU) activation function is applied. This function introduces non-linearity by turning all negative values into zero, helping the model to learn complex and useful patterns.

Once features are extracted, the data moves through pooling layers, which shrink the size of the feature maps. This step reduces the number of parameters, helping to prevent overfitting and making the model more resilient to changes in the input image, like rotations or shifts. The features are then flattened into a single one-dimensional vector and sent to fully connected layers. These layers bring together all the learned features to generate predictions. In the final step, a SoftMax activation function is applied to turn the output values into probability scores. In this case, the model outputs probabilities for three ethnic groups: Malay (0.7), Chinese (0.2), and Indian (0.1). The highest probability indicates the predicted ethnicity.

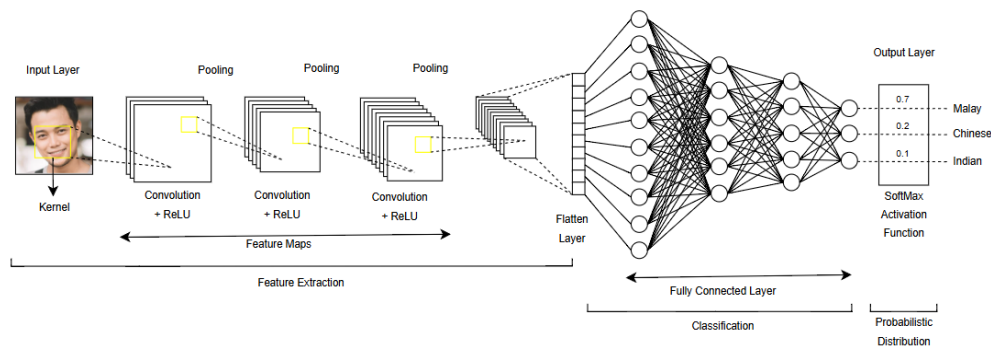


Figure 1. CNN diagram architecture

The VGG-Face model, a variant of the VGG-16, were pre-trained on a large dataset of facial images, making it effective at extracting critical facial features such as the eyes, nose, and mouth which is the key elements for distinguishing ethnicities (Nakada et al., 2017). By leveraging transfer learning, this study avoided the need to train a model from scratch, saving significant computational resources and time. Most layers were frozen to retain general facial feature extraction while minimizing overfitting, and the original classification head was replaced with a task-specific one comprising a fully connected layer, dropout, and a softmax output for the three target ethnicities: Malay, Chinese, and Indian. A few higher-level layers were unfrozen and fine-tuned on the dataset to help the model adapt to ethnicity-specific facial variations.

The VGG-Face model comprises 37 layers grouped into six blocks, each designed to extract facial features with increasing complexity. The initial two blocks are designed to capture basic features like edges and textures. They include two convolutional layers with ReLU activation functions, followed by max-pooling layers to reduce the spatial size while retaining important details. The following three blocks are aimed at learning more complex features, such as facial patterns and structures. Each of these blocks contains three convolutional layers with ReLU activations and max pooling. This layered structure helps the model gradually learn more abstract and distinguishing features. In the final stage, the feature maps are flattened and passed through three fully connected

layers, followed by a SoftMax layer, which combines all the learned information to make accurate classifications.

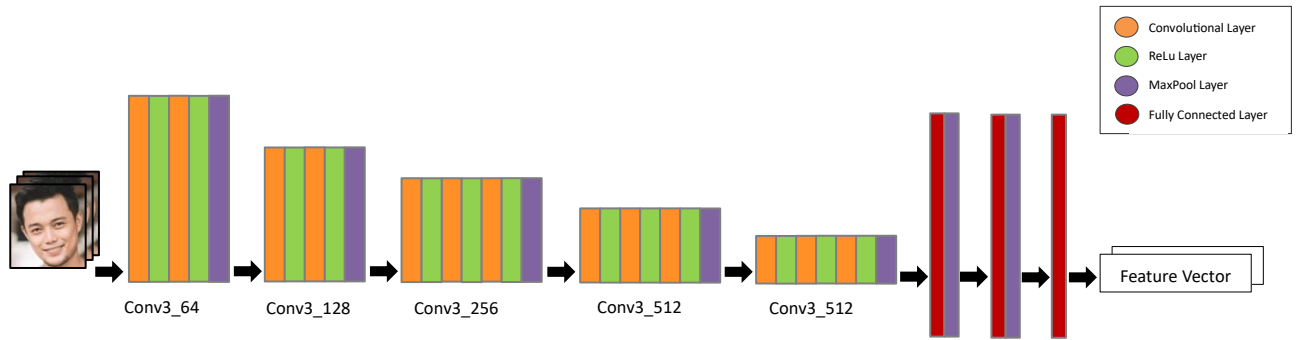


Figure 2. VGG Face Network Architecture

Model performance is generally assessed using metrics like accuracy, precision, recall, and F1 score. Although accuracy is widely used, it may not provide a reliable evaluation when dealing with imbalanced datasets (Hossin and Sulaiman, 2015). In such cases, precision, recall, and the F1 score offer a more comprehensive and balanced measure of performance, especially in the presence of uneven class distributions. Confusion matrices are also useful for visualizing misclassifications and identifying areas for improvement.

MODEL IMPLEMENTATION

Hardware Specification

The model was developed and trained on Google Colab, leveraging cloud-based hardware for deep learning. Table 3 provides an overview of the hardware and software configurations used in this project.

Table 3. Overview of Hardware and Software Specifications

Component	Specification
GPU	T4 GPU (Free), NVIDIA L4 GPU 22.5 GB VRAM (Pro)
RAM	12.7 GB (T4 GPU), 53.0 GB (L4 GPU)
Operating System	Ubuntu (via Google Colab)
Environment	Google Colab, Jupyter Notebook
Libraries and Framework	TensorFlow/Keras, OpenCV, NumPy, Matplotlib

Dataset Collection

The dataset included facial images of three ethnic groups: Malay, Chinese, and Indian, sourced primarily via web scraping from Google Images and websites using Selenium-based Python scripts. Manual selection of public celebrity and athlete images supplemented the dataset, with an additional Indian dataset from a GitHub repository (Katti & Arun, 2019).

Preprocessing was carried out to ensure dataset quality and compatibility with the model. Low-quality or blurry images were removed using Laplacian variance, and duplicate images were eliminated. MTCNN was applied for accurate face detection and cropping, followed by manual verification to confirm image relevance. The dataset was balanced to include 460 images per ethnic group, totaling 1,380 images. All images were resized to 224×224 pixels, converted to NumPy arrays, reshaped into a 4D tensor, and normalized to a [0,1] range. Ethnicity labels were also encoded as integers for model training. Table 4 shows the summary for data collection and preprocessing. Then, the dataset was split into three subsets. Initially, 80% of the data (1,104 images) was allocated for training and 20% (276 images) for testing. From the training portion, 12% (133 images) was set aside for validation, leaving 68% (971 images) as the final training set. This split ensured the model had sufficient data to learn from, while also allowing for performance monitoring during training and unbiased evaluation after training.

Table 4. Data Collection and Preprocessing Summary

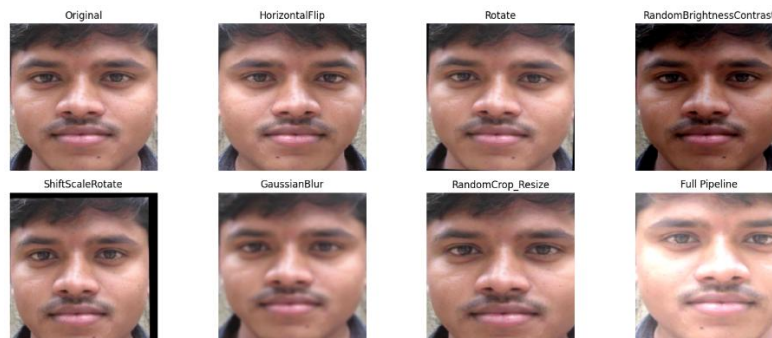
Ethnic Group	Initial Number of Images	Images After Cleaning	Encoded Integer
Indian	528	460	1
Malay	553	460	2
Chinese	570	460	3

Data Augmentation

Data augmentation was applied using the Albumentations library for better model generalization and prevent overfitting. The transformation used was summarized in Table 5. These augmentations introduced variability in lighting, orientation, and scale which help the model learn more robust, invariant features across different real-world conditions.

Table 5. Augmentation Pipeline

Transformation	Limit	Probability (p)
Horizontal Flip	-	0.5
Rotate	15° (Fine-Tuned 1), 10° (Fine-Tuned 2)	0.3
Random Brightness/Contrast	brightness = 0.2, contrast = 0.2	0.3
Shift, Scale, Rotate	shift = 0.1, scale = 0.1, rotate = 10	0.3
Gaussian Blur	blur = (3, 7)	0.2
Random Crop	height = 200, width = 200	0.2

**Figure 3.** Visualization of Data Augmentation

Model Training

An initial model was trained from scratch without any pre-trained weights for evaluating the benefits of transfer learning. The model was trained using default settings without data augmentation, relying solely on the available dataset. Subsequently, three models were trained using transfer learning, starting from a pre-trained architecture and fine-tuning it on the ethnicity classification task:

1. Baseline: Applied L2 regularization of 0.01 and dropout rate of 0.5, without data augmentation.
2. Fine-Tuned 1: L2 regularization of 0.001, dropout rate of 0.3, and augmentation Pipeline 1.
3. Fine-Tuned 2: L2 regularization of 0.0001, dropout rate of 0.35, and augmentation Pipeline 2.

The model training was carefully configured to balance learning efficiency and overfitting prevention. Table 6 summarizes the key parameters and techniques used during training across all models, including optimization strategy, and regularization.

Table 6. Training Configuration Summary

Configuration	Description
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Loss Function	Categorical cross-entropy
Optimizer	Adam optimizer
Epochs	1000
Batch Size	256
Early Stopping (Patience)	30 epochs
Checkpointing	Save model weights
ReduceLROnPlateau	Adjusts learning rate when validation loss plateaus (factor=0.5, patience=10, min_lr=0.000001).

RESULT AND DISCUSSION

All of the three models performance was evaluated using the validation and test sets, with results summarized in Table 7. The metrics include validation loss, test loss, validation accuracy, test accuracy, robustness check, and the actual loss difference. All models met the robustness criterion, with the no transfer learning showing the smallest loss difference but with less accuracy. Fine-Tuned 2 with larger gap (0.0420) is acceptable given its lower losses and higher accuracies. The choice of L2 regularization (0.0001 for Fine-Tuned 2 versus 0.001 for Fine-Tuned 1) and dropout (0.35 versus 0.3) influenced performance. Fine-Tuned 2's lighter regularization allowed better convergence, while the increased batch size (256) and weight decay (0.0001) stabilized training.

Table 7. Performance Comparison of Models

Model	Val Loss	Test Loss	Val Accuracy	Test Accuracy	Loss Difference – Robust Check (Val - Test Loss < 0.05)
No Transfer Learning	1.1035	1.1007	33.08%	33.33%	0.0032
Baseline	0.6561	0.6794	89.47%	88.41%	0.0168
Fine-Tuned 1	0.4226	0.4345	91.73%	89.86%	0.0276
Fine-Tuned 2	0.3313	0.3501	91.73%	91.30%	0.0420

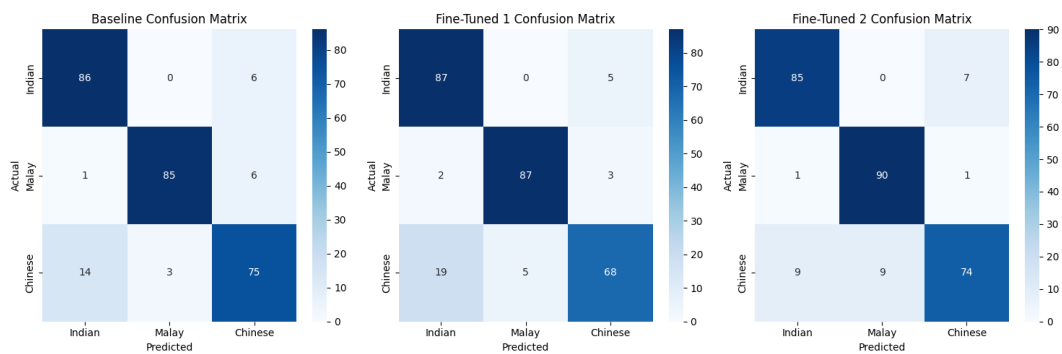
Transfer learning significantly improved performance, as evidenced by the superior results of Fine-Tuned 1 and Fine-Tuned 2 over the Baseline. The pre-trained VGG-Face weights enabled the models to leverage general facial features, while fine-tuning the last 6 layers adapted the network to ethnicity-specific patterns. Fine-Tuned 2's lower regularization (L2=0.0001) and moderate augmentation pipeline resulted in the best performance, indicating an optimal balance of model capacity and generalization. The Baseline's higher losses and lower accuracies highlight the limitations of training without augmentation and with default regularization, underscoring the effectiveness of transfer learning combined with fine-tuning.

The use of augmentation in Fine-Tuned 1 and Fine-Tuned 2 reduced overfitting, as seen in the lower validation losses and smaller generalization gaps compared to the Baseline. The moderated augmentation in Fine-Tuned 2 (e.g., reduced rotation limit) likely contributed to its superior performance by introducing variability without overwhelming the model.

The classification reports provide precision, recall, and F1-score for each ethnicity class. Fine-Tuned 2's high test accuracy (91.30%) suggests balanced performance across classes, but the confusion matrices (Figure 4) should be inspected to identify any specific misclassifications. For instance, if certain ethnicities are consistently confused, this could indicate dataset bias or insufficient distinguishing features. The high validation and test accuracies of Fine-Tuned 2 and Fine-Tuned 1 (both 91.73% on validation) suggest robust class-wise performance, but further analysis of per-class metrics is recommended to ensure fairness.

Table 8. Classification Performance Summary for Test Set

Model	Ethnicity	Precision	Recall	F1-Score
Baseline	Indian	0.85	0.93	0.89
	Malay	0.97	0.92	0.94
	Chinese	0.86	0.82	0.84
Fine-Tuned 1	Indian	0.81	0.95	0.87
	Malay	0.95	0.95	0.95
	Chinese	0.89	0.74	0.81
Fine-Tuned 2	Indian	0.89	0.92	0.91
	Malay	0.91	0.98	0.94
	Chinese	0.90	0.80	0.85

**Figure 4.** Confusion Matrices

LIMITATION AND FUTURE WORK

Despite promising results, the study faced limitations, primarily the small dataset, which restricted generalizability. Though data augmentation helped, the dataset lacked intra-class diversity in aspects like facial structure, lighting, and expression. Additionally, many images were sourced from Google and featured celebrities, potentially introducing bias due to image enhancements or unnatural conditions.

While this study focused on classifying the three major Malaysian ethnic groups, the country's rich ethnic diversity, especially among the Bumiputera communities of Sabah and Sarawak remains underrepresented. Future work should aim to expand the dataset to include these underrepresented groups to better reflect the full spectrum of Malaysian ethnic diversity. This would involve collecting a larger, more inclusive, and demographically balanced dataset, potentially through collaboration with local institutions, community outreach, or publicly available databases with appropriate consent. By incorporating these additional groups, future models could provide a more comprehensive and representative classification system, useful for applications in demographic research, public policy, and inclusive AI systems. Careful attention must also be paid to ethical considerations, ensuring fairness, privacy, and the avoidance of cultural stereotyping or misuse.

CONCLUSION

This project successfully demonstrated the viability of using transfer learning with a VGG-Face model for ethnicity classification in the Malaysian context. The approach achieved solid performance despite data limitations, showing the potential of deep learning in demographic classification tasks. With improved data diversity and model architectures, future work can build on this foundation to develop more accurate, fair, and ethically responsible systems for real-world applications.

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REFERENCES

- Alzubaidi, L., Zhang, J., Humaidi, A. J., Al-Dujaili, A., Duan, Y., Al-Shamma, O., Santamaría, J., Fadhel, M. A., Al-Amidie, M., & Farhan, L. (2021). Review of deep learning: concepts, CNN architectures, challenges, applications, future directions. *Journal of Big Data*, 8(1). <https://doi.org/10.1186/s40537-021-00444-8>
- Ahmed, M. A., Choudhury, R. D., & Kashyap, K. (2020). Race estimation with deep networks. *Journal of King Saud University - Computer and Information Sciences*, 34(7), 4579–4591. <https://doi.org/10.1016/j.jksuci.2020.11.029>
- Ache, A. I., Kassir, M. M., & Ibrahim, H. M. (2023). Ethnicity Classification: Discriminating between Iranian and Asian populations using hybrid deep learning algorithm. *Journal of Image Processing and Intelligent Remote Sensing*, 41, 20–33. <https://doi.org/10.55529/jipirs.41.20.33>
- Alghaili, M., AlBdairi, A. J., Li, Z., & Katiyalath, M. (2023). Generating embedding features using deep learning for ethnics recognition. *The International Arab Journal of Information Technology*, 20(4). <https://doi.org/10.34028/iajit/20/4/13>
- Chicho, B. T., & Sallow, A. B. (2021). A comprehensive survey of deep learning models based on KERAS framework. *Journal of Soft Computing and Data Mining*, 2(2). <https://doi.org/10.30880/jscdm.2021.02.02.005>
- Hossin, M., & Sulaiman, MN. (2015). A review on Evaluation Metrics for Data Classification Evaluations. *International Journal of Data Mining & Knowledge Management Process*, 5(2), 01–11. <https://doi.org/10.5121/ijdkp.2015.5201>
- Jahan, S. T., & Mustafa, R. (2024). Ethnicity Detection with Convolutional Neural Network (CNN): Bangladesh Perspectives. *American Journal of Interdisciplinary Research and Innovation*, 5–12. <https://doi.org/10.54536/ajiri.v3i4.3772>
- Jilani, S. K., Ugail, H., Bukar, A. M., & Logan, A. (2019). *On the Ethnic Classification of Pakistani Face using Deep Learning* (Vol. 72, pp. 191–198). <https://doi.org/10.1109/cw.2019.00039>
- Kalkatawi, A., & Saeed, U. (2024). Ethnicity Classification Based on Facial Images using Deep Learning Approach. *International Journal of Advanced Computer Science and Applications*, 15(2). <https://doi.org/10.14569/ijacsa.2024.0150223>
- Katti, H., & Arun, S. P. (2019). Are you from North or South India? A hard face-classification task reveals systematic representational differences between humans and machines. *Journal of Vision*, 19(7), 1. <https://doi.org/10.1167/19.7.1>
- Makolo, A. U., & Dada, O. V. (2023). Ethnicity Classification: a Machine Learning approach. *Zenodo (CERN European Organization for Nuclear Research)*. <https://doi.org/10.5281/zenodo.7655148>
- Nakada, M., Wang, H., & Terzopoulos, D. (2017). *AcFR: Active Face Recognition Using Convolutional Neural Networks*. <https://doi.org/10.1109/cvprw.2017.11>
- Novac, O., Chirodea, M. C., Novac, C. M., Bizon, N., Oproescu, M., Stan, O. P., & Gordan, C. E. (2022). Analysis of the application efficiency of TensorFlow and PyTorch in convolutional neural network. *Sensors*, 22(22), 8872. <https://doi.org/10.3390/s22228872>
- Vo, T., Nguyen, T., & Le, C. T. (2018). Race recognition using deep convolutional neural networks. *Symmetry*, 10(11), 564. <https://doi.org/10.3390/sym10110564>
- Wang, W., He, F., & Zhao, Q. (2016). Facial Ethnicity Classification with Deep Convolutional Neural Networks. In *Lecture notes in computer science* (pp. 176–185). https://doi.org/10.1007/978-3-319-46654-5_20
- Yuan, J. (2024). Performance analysis of deep learning algorithms implemented using PyTorch in image recognition. *Procedia Computer Science*, 247, 61–69. <https://doi.org/10.1016/j.procs.2024.10.008>